

The long-term effects of drinking alcohol for stress coping on mental health symptoms among Fukushima nuclear plant workers: The Fukushima NEWS Project Study

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Abstract

Objective

The Fukushima nuclear disaster in 2011 caused psychological distress among plant workers. A three-year follow-up study was conducted to understand the longitudinal relationship between workers' alcohol drinking for stress coping and mental health symptoms.

Methods

The present study analyzed 405 Fukushima Daiichi and Daini nuclear power plant workers, who responded to the open-ended questions about their own stress reduction efforts in 2012. Elements were extracted from the descriptions and categorized into 'coping by drinking alcohol' and 'adaptive stress coping'. Mental health symptoms were assessed annually from 2012 to 2014. The relationships between stress coping and mental health symptoms over time were analyzed using a mixed-effects logistic regression model.

Results

Thirty-five workers drank alcohol for stress coping (12 drank alcohol only; 23 reported adaptive stress coping as well). The workers who drank alcohol only had significantly high Impact of the Event Scale-Revised (IES-R) ($\beta = 9.61$, $p = 0.008$), Athens Insomnia Scale (AIS) ($\beta = 2.66$, $p = 0.008$), and CAGE scores ($\beta = 1.38$, $p < 0.001$) in 2012. The significant effects on IES-R and CAGE persisted through 2014; however, the effect on AIS decreased over time ($\beta = -1.16$, $p = 0.045$). The workers who drank alcohol and implemented adaptive stress coping had significantly higher CAGE scores in 2012 ($\beta = -0.08$, $p = 0.008$), and the impact increased over time ($\beta = 0.51$, $p = 0.003$).

Conclusions

In the study population, drinking alcohol for stress coping increased the risk of problem drinking, as well as post-traumatic stress response and insomnia, especially in the absence of adaptive stress coping. Because these effects persist over years, professional education and guidance and continuous monitoring of coping content are considered necessary from the early post-disaster period.

INTRODUCTION

On March 11, 2011, a tsunami following an earthquake with a magnitude of 9.0 heavily damaged the northern coast of Japan. This event led to a nuclear disaster at the Tokyo Electric Power Company (TEPCO) Fukushima Daiichi Nuclear Power Plant (Daiichi plant). The disaster involved a critical core meltdown and the emission of radioactive materials into the atmosphere, which put the workers' lives at risk from high radiation exposure and explosions [1]. This nuclear power plant disaster was the second worst in history, following the 1986 Chornobyl (Chernobyl) disaster [2]. The TEPCO Fukushima Daini Nuclear Power Plant (Daini plant), located 12 kilometers away, was also damaged by the tsunami, but did not experience core damage [1].

Some previous studies have examined the impact of disasters on mental health consequences, including depression and post-traumatic stress disorder (PTSD) symptoms [3, 4]. Goldmann and Galea also suggested that human-made technological disasters tend to have a more pronounced psychological impact than natural disasters [5]. Similar trends were observed in Fukushima, where cases of PTSD, depression, anxiety, and general psychological distress (GPD) were reported [6]

According to Lazarus and Folkman's psychological stress theory, a potentially stressful event (stressor) is first evaluated as irrelevant, harmless, positive, or stressful, followed by a secondary evaluation of how threatening it is, how it affects us, and whether we can cope with it. Then, a response is made to the stressor [7]. A validation study of the factorial validity of Lyne and Roger's coping reported that it is appropriate to classify coping strategies into three categories: rationale or active coping, emotion-focused coping, and avoidance [8]. Avoidance coping is evading stressors through distraction, denial, or withdrawal behaviors (e.g., procrastinating or distracting oneself with entertainment instead of facing stress, and using alcohol or substances to escape from stressful situations) [8]. As an aspect of post-disaster coping, increased alcohol drinking, considered as negative coping, was significantly correlated with depression and poor quality of life (QOL) among the Hurricane Katrina survivors [9]. A survey of those directly exposed to the 9/11 terrorist attacks on the World Trade Center (WTC) found that intentional self-medication with alcohol (ISMA) was significantly associated with the risk of PTSD [10].

In Chornobyl (Chernobyl), alcohol consumption was found to affect mental health [11] and was also suggested to contribute to increased mortality [12]. In the Fukushima Nuclear Power Plant, we reported that younger age at the time of the disaster and specific disaster-related experiences significantly increased alcohol consumption after the disaster [13]. Furthermore, certain disaster experiences were significantly associated with long-term symptoms of problematic drinking [14]. However, there are no previous studies on the effects of alcohol-related coping behaviors on mental health among post-disaster nuclear power plant workers.

In this study, we focused on the use of alcohol consumption as a stress coping strategy to examine its long-term effects on mental health. We hypothesized that nuclear power plant workers who drank alcohol as a stress coping strategy would have worse mental health or delayed recovery of mental health symptoms compared to those who used other adaptive stress coping methods after the disaster.

METHODS

Full-time TEPCO workers from the Daiichi plant ($n = 1,053$) and the nearby Daini ($n = 707$) plant participated in the Fukushima Nuclear Energy Workers' Support (NEWS) Project study, a longitudinal research project assessing the health effects of nuclear disaster among the front-line workers. Full details of the project have been described elsewhere.^{6–8} Eighty-five percent of the plant workers (1,495 individuals) responded after enrollment with informed consent. The present study was approved by the ethics committees of Juntendo University (No. 2020053), the National Defense Medical College (No. KAN-61), and Mejiro University (No. 20-I-015).

The Fukushima Daiichi and Daini plant workers responded to a series of self-reported questionnaires, which included questions about demographics and disaster-related experiences at the time of the disaster in 2011 [15]. They were asked to "Please describe any creative ways through which you try to relieve stress at work or at home" in the follow-up self-reported questionnaire and responded with free-text answers in 2012 (May–June, 14–15 months post-disaster). They also responded to a follow-up self-reported questionnaire on mental health symptoms (PTSD, depression, insomnia, and problem drinking) in 2012, 2013 (November, 32 months post-disaster), and 2014 (November, 44 months post-disaster). Of the respondents, 388 workers were excluded due to missing baseline information, such as age or sex, or missing mental health questionnaire answers in 2012. A total of 1,107 individuals were included in the preliminary study (405 individuals with reported stress coping, and 705 individuals without any reported stress coping). Since our main analysis was to determine the longitudinal relationship between alcohol drinking as a stress coping strategy and mental health among the nuclear power plant workers, we only included 405 workers who provided answers to the stress coping and mental health symptom questions in 2012..

In the analytic sample, 177 (43.7%) responded at all three time points, 142 (35.1%) responded at two of three time points, and the remaining 86 (21.2%) responded at only one of the three time points. A total of 901 observations were included

(405 in 2012, 253 in 2013, and 243 in 2014).

We gathered workers' socio-demographic information and disaster-related experiences in the form of the Impact of the Event Scale-Revised (IES-R), the Kessler Psychological Distress Scale (K6), and the Athens Insomnia Scale (AIS) in 2011, two to three months after the disaster. IES-R, K6, AIS, and CAGE were collected annually via self-reported questionnaires from 2012 to 2014. Disaster-related experiences (Table 1) were dichotomously coded as "yes" or "no" and chosen based on the stressors encountered by the nuclear plant workers as described during their first medical interview post-disaster, which was conducted prior to baseline data collection by Tanigawa, T (a co-author; April 16–19, 2011). Experiencing discrimination/slurs was identified and included as a major stressor because nuclear plant workers were blamed severely for the nuclear contamination and home evacuations following the disaster.

Table 1
Descriptive statistics of analysis sample at the time of the disaster in 2011

Numbers of subjects, N	405
Mean age, years	41.2
Male, %	93.3
Daiichi plant worker, %	41.7
Pre-existing psychiatric illness, %	2.0
Disaster-related experiences	
- The experience of life-threatening danger, %	45.7
- Major property loss, %	30.4
- Experiences of discrimination/slurs, %	14.1
- The experience of escaping from a tsunami, %	13.1
- The witnessing of plant explosion, %	27.7
- The death of a family member, %	6.4
- The death of a colleague, %	20.7
- Home evacuation, %	68.2

The workers were asked to describe their own stress coping in the open-ended question: "Please describe any ways through which you cope with stress at work or at home." Those workers describing any type of stress coping behavior (excluding such answers as "none" or "nothing in particular") counted as having implemented stress coping. If the workers' stress coping was associated with alcohol drinking, it was counted as implementing stress coping associated with alcohol drinking or counted as having adaptive stress coping. Full elements of stress coping extracted from the free-text responses are listed in Appendix 3.

The Japanese version of the IES-R was used for PTSR assessment [16]. The IES-R contains 22 items to assess the psychological impact of potentially traumatic events in three clusters—intrusion, avoidance, and hyperarousal. The Diagnostic and Statistical Manual, Fifth Edition, Text Revision (DSM-5-TR) contains four symptom clusters: the aforementioned three clusters and negative alterations in cognition and mood [17]. However, a Japanese version of the DSM-5-TR was unavailable at the time of this study. Each question is scored on a 5-point scale (0–4), with total scores ranging from 0 to 88. Scores of 25 or higher indicate significant PTSRs [17].

The Japanese version of the K6 [18] was used to measure GPD. The K6 is a well-established self-reported questionnaire used for mood and anxiety disorders, as defined by the Diagnostic and Statistical Manual, fourth edition (DSM-IV) [19, 20]. The K6 uses six items to assess feelings of nervousness, hopelessness, restlessness or fidgeting, depression, worthlessness, and everything that required effort in the past 30 days. Each item is scored on a 5-point scale (0–4), with total scores ranging from 0 to 24. Scores of 13 or higher indicate serious mental illness [21].

We measured insomnia by the AIS, which contains eight questions regarding difficulty initiating sleep (DIS), difficulty maintaining sleep (DMS), and early morning awakening (EMA) [22]. We scored each AIS item on a 4-point Likert scale ranging from 0 to 3, and a total score of 6 or higher was indicative of clinical insomnia [22]. In the context of this study, we restricted our attention to three symptoms that might characterize different patterns of insomnia, which were assessed by the first three questions of the AIS: DIS, DMS, and EMA.

Problem drinking was measured with four questions from the CAGE questionnaire, a self-reported scale widely used to screen for problem drinking in the general population [23]. The four items involved questions regarding cutting down on drinking (C), annoyance over criticism (A), guilty feelings (G), and eye-openers (E). The participants answered yes or no to the following questions regarding their current drinking: “Have you felt that you should cut down on your drinking?”; “Have people annoyed you by criticizing your drinking?”; “Have you felt bad or guilty about your drinking?”; and “Have you had a drink first thing in the morning to steady your nerves or get rid of a hangover (eye-opener)?” A Japanese translation of these questions was used [24]. Items were scored as 0 or 1, with the total score ranging from 0 to 4 and higher scores representing greater levels of problem drinking. Scores of 2 or higher indicated clinically significant problem drinking [23].

Statistical Analysis

All analyses were conducted using SAS version 9.4 (SAS Institute Inc., Cary, NC, USA). The linear trend of mental health scores (IES-R, K6, AIS, and CAGE) over time was tested using the Wald test for the coefficient associated with time and mixed-effect longitudinal beta regression models [25].

To examine the association between drinking alcohol for stress coping (drinking alcohol only or drinking alcohol with adaptive stress coping) and mental health scores (IES-R, K6, AIS, and CAGE), we used the following two models:

Model 1:

$$Y_{ij} = \beta_0 + \beta_C \text{Alcohol}_i + \beta_T \text{Time}_{ij} + (\text{Effect of other covariates}) + b_i$$

Model 2:

$$Y_{ij} = \beta_0 + \beta_C \text{Alcohol}_i + \beta_T \text{Time}_{ij} + \beta_{CT} (\text{Alcohol}_i * \text{Time}_{ij}) + (\text{Effect of other covariates}) + b_i$$

where Y_{ij} is a continuous outcome variable of mental health scores (IES-R, K6, AIS, and CAGE) in survey wave j ($j = 1, 2$, or 3) and 0 if otherwise; Alcohol_i is a binary variable taking the value of 1 if individual i reported drinking alcohol for stress coping (drinking alcohol only or drinking alcohol with adaptive stress coping); Time_{ij} represents the time at completion of the questionnaire, measured in years since the disaster (taking values from 1 to 3); and b_i is a normally distributed subject-specific random effect. The models were further adjusted for a vector of covariates assessed at the time of the disaster in 2011: age, gender, and job location (Daiichi or Daini plant). Model 1 assumed that exposure to the disaster-related event had a constant effect on the anxiety score over time, whereas Model 2 allowed the effect to vary linearly (on the logarithmic scale) with time. SAS Proc MIXED was used to fit the mixed-effect longitudinal beta regression models.

Because gender and pre-existing psychiatric illness may act as moderating variables in the association between disaster-related experiences and anxiety, we performed two sensitivity analyses: one restricting the study population to male workers and the other restricting it to workers without pre-existing psychiatric illness.

RESULTS

We performed sensitivity analyses of the difference in demographics, mental health status and the changes over time between the present sample with reported stress coping (N = 405) and the preliminary study population including the workers without reported (N = 1,107). There was no major difference in the demographics and disaster-related experiences (Appendix 1) and longitudinal changes in K6, AIS, and CAGE (Appendix 2) between this study population and the overall population including the workers without reported stress coping, but the IES-R score significantly increased over the years (model slope: 0.23, $p < 0.001$) only in the overall population including the workers without reported stress coping (Appendix 2)

The basic characteristics of the participants at the time of the disaster in 2011 are presented in Table 1. The mean age of the study population with reported stress coping was 41.4 years old in 2011. Almost all (93.3%) of the individuals in our sample were men, 41.7% of them worked at the Daiichi Nuclear Power Plant, and only 2.0% of the workers had a pre-existing psychiatric illness.

Table 2 shows the workers' stress coping behavior reported at baseline in 2012. Twelve workers (3.0%) reported drinking alcohol only for stress coping. Twenty-three workers (5.7%) reported both alcohol drinking and adaptive stress coping. The other 370 workers (91.4%) reported adaptive stress coping only.

Table 2
Types of reported self-coping

	N	%
Subjects with reported stress coping	405	100
(I) Drinking alcohol only	12	3.0
(II) Drinking alcohol with adaptive stress coping	23	5.7
(III) Adaptive stress coping only	370	91.4

Table 3 shows the changes in mental health scores. The IES-R score did not have significant changes over the years (model slope: 0.46, $p = 0.25$); however, there were significant increases in the K6, AIS, and CAGE scores over the years (K6 model slope: 0.60, $p < 0.001$; AIS model slope: 0.48, $p < 0.001$; CAGE model slope: 0.08, $p = 0.027$).

Table 3
Trends of the IES-R, K6, AIS, and CAGE scores (405 individuals)

	Year of survey											
	2012			2013			2014			<i>p</i> for trend		
(A) IES-R												
Response rates of workers followed at each year, %	405			100%			250			61.7%	241	59.5%
IES-R score, mean ± SD	13.0	±	12.6	14.0	±	13.4	13.7	±	14.5	0.25 ^a		
IES-R ≥ 25, %	17.0			20.0			20.8			0.084 ^b		
(B) K6												
Response rates of workers followed at each year, %	405			100%			253			62.5%	243	60.0%
K6 score, mean ± SD	4.2	±	3.9	5.0	±	4.7	5.4	±	5.3	< 0.001 ^a		
K6 ≥ 13, %	3.7			8.7			12.8			< 0.001 ^b		
(C) AIS												
Response rates of workers followed at each year, %	405			100%			252			62.2%	243	60.0%
AIS score, mean ± SD	5.0	±	3.4	6.0	±	3.9	5.8	±	3.9	< 0.001 ^a		
AIS ≥ 6, %	40.3			54.4			49.0			0.001 ^b		
(D) CAGE												
Response rates of workers followed at each year, %	405			100%			253			62.5%	243	60.0%
CAGE score, mean ± SD	0.5		0.8	0.7	±	1.0	0.7	±	1.0	0.027 ^a		
CAGE ≥ 2, %	12.1			17.0			18.9			0.007 ^b		
^a Values based on the Wald test for the coefficient associated with time in a mixed-effect longitudinal beta regression model												
^b Values based on the Wald test for the coefficient associated with time in a mixed-effect logistic regression model												

Table 4 summarizes the results from linear mixed-effects models describing the effect of drinking alcohol for stress coping reported at baseline in 2012 (drinking alcohol only or drinking alcohol with adaptive stress coping vs. adaptive stress coping only). The PTSR, AIS, and CAGE scores were significantly high at baseline in 2012 among the workers who reported drinking alcohol only for stress coping (IES-R: $\beta = 9.61$, $p = 0.008$; AIS: $\beta = 2.66$, $p = 0.008$; CAGE: $\beta = 1.38$, $p < 0.001$). The interactions of time with the association between drinking alcohol only for stress coping and IES-R and CAGE were not significant; therefore, these associations remained significant for two years. However, the association between drinking alcohol only for stress coping and AIS significantly decreased over time ($\beta = -1.16$, $p = 0.045$). The CAGE score was also significantly high at baseline in 2012 among the workers who reported drinking alcohol and adaptive stress coping ($\beta = 0.08$, $p = 0.006$), and this association significantly increased over time ($\beta = 0.51$, $p = 0.003$). The workers' alcohol drinking with adaptive stress coping had no significant impact on the association with IES-R, K6, and AIS. The

trends were also not attenuated during sensitivity analyses upon restricting the study population to male workers or to workers without pre-existing psychiatric illness (Appendix 4 and 5). However, in the workers without pre-existing psychiatric illness, although an increase in problem drinking due to alcohol drinking for stress coping was observed at baseline, no significant increase in longitudinal changes was found (Appendix 5).

Table 4
Multivariate-adjusted β estimates for the IES-R, K6, AIS, and CAGE scores associated with alcohol consumption in 2012 (405 individuals)^a

	(I) Drinking alcohol only (N = 12) vs. (III) Adaptive stress coping only (N = 370)						(II) Drinking alcohol with adaptive stress coping (N = 23) vs. (III) Adaptive stress coping only (N = 370)					
	Effect in 2012			Time-dependent effect (for 1-year increment)			Effect in 2012			Time-dependent effect (for 1-year increment)		
	B	(S.E.)	P	β	(S.E.)	P	β	(S.E.)	P	β	(S.E.)	P
IES-R score												
Multivariate-adjusted	9.61	(3.63)	0.008	-1.63	(2.19)	0.46	0.74	(2.59)	0.77	-0.24	(1.90)	0.90
K6 score												
Multivariate-adjusted	2.14	(1.13)	0.059	-1.30	(0.77)	0.093	0.64	(0.81)	0.43	0.04	(0.70)	0.95
AIS score												
Multivariate-adjusted	2.66	(0.99)	0.008	-1.16	(0.58)	0.045	-0.15	(0.73)	0.84	-0.16	(0.52)	0.75
CAGE score												
Multivariate-adjusted	1.38	(0.23)	< 0.001	-0.24	(0.15)	0.11	0.08	(0.03)	0.006	0.51	(0.17)	0.003
^a Adjusted for baseline age, gender, and job location at the time of the disaster in 2011												

DISCUSSION

To our knowledge, this is the first study to examine the longitudinal relationships between alcohol drinking, stress coping, and the mental health symptoms of the nuclear plant workers after a major nuclear disaster. The workers who drank alcohol only for stress coping had symptoms of PTSD, insomnia, and problem drinking in 2012, and these effects on PTSD and problem drinking persisted through 2014; however, the effect on insomnia decreased over time. The workers with alcohol consumption and adaptive stress coping had symptoms of problem drinking, and the impact increased over time.

Our finding that drinking alcohol as a stress coping strategy was significantly associated with PTSD is similar to the results reported in studies of people exposed to the 9/11 disaster [10]. Combined use with active stress coping reduced the negative effects of drinking alcohol on PTSD in this study. This result was supported by two previous reports; “acceptance” as an early coping behavior significantly reduced the PTSD (OR = 0.71; 95% CI: 0.56–0.89) related to the disaster, at four months after the 9/11 terrorist attack [26], and PTSD was reduced by positive religious coping (OR = 0.49; 95% CI: 0.30–0.79) among the victims after Hurricane Katrina [27].

Drinking alcohol only for stress coping significantly affected long-term insomnia, and the impact was reduced if adaptive stress coping was also added to the coping strategies in the present study. Significant research has been conducted on

the effects of alcohol on sleep. Drinking alcohol produces changes in the first half of sleep, such as a shortening of the time that it takes to fall asleep, an increase in slow wave sleep (deep sleep), and a decrease in REM sleep (sleep in which the body rests but the brain is active) [28]. However, the effects are not sustained, and in later sleep phases, there is a decrease in slow wave sleep, an increase in REM sleep, and an increase in mid-wake (waking up in the middle of the night and taking longer to fall asleep again), regardless of the amount of alcohol, of gender, or age [28, 29]. It has also been found that alcohol worsens daytime sleepiness [29] and causes the onset and worsening of symptoms of obstructive sleep apnea [30]. As such, alcohol disrupts sleep architecture and results in poor sleep quality. On the other hand, it has been reported that active coping is negatively associated with insomnia severity [31]. The results of this study—combined use of alcohol and adaptive stress coping did not significantly elevate insomnia but coping only with alcohol drinking significantly elevated insomnia—suggest that although insomnia can be caused by alcohol consumption, it may be reduced by adaptive stress coping.

The present study showed that the use of alcohol consumption for stress coping significantly affected problem drinking in subsequent years. However, the increase over time was not significant if eight workers with pre-existing psychiatric illness at the time of the disaster in 2011 were excluded. These results suggest that the increase over time may be due to the effect of those with a history of psychiatric illness at the time of the disaster. In a study of Hurricane Katrina victims, positive religious coping significantly reduced the risk of alcohol use (OR = 0.13; 95% CI: 0.05–0.33) [19]. The Hurricane Katrina study [27] is not directly comparable to the present results because it focused only on positive religious coping and the amount of drinking, not problem drinking. However, the present study showed a strong association between drinking alcohol for stress coping and problem drinking symptoms. We also believe that it is important to understand the pre-existing psychiatric illness of the affected population, since the results suggest that the risk of problem drinking may increase over time for those with a history of mental illness.

Unlike the other three items studied, GPD was not affected by alcohol consumption for stress coping in the current study. It is possible that this may be the result of a cancellation of the negative psychophysical effects by the positive effects in terms of mood improvement and better connections with the surrounding parties through drinking opportunities [27].

From July 2011 to April 2012, on-site psychiatrists and clinical psychologists provided mental health support, including clinical treatment and continuous counseling, to a limited number of workers with severe work-related trauma, distress, and experiences of grief [32]. Of these, 85 workers (6%) attended more than three treatment or counseling visits by 2014. Participation in such sessions over time is expected to have a small effect on mental health symptoms among the plant workers. Nevertheless, our results point to the persistence of mental health symptoms associated with the use of alcohol consumption for stress coping.

There are several limitations to the present study. First, we used self-report, open-ended questionnaires to gather details about stress coping. It is possible that the presence or absence of reports and the degree of detail differed due to different interpretations by each worker. Future research should include validation with surveys that include structured questionnaires and/or interviews. However, since a detailed factor decomposition was conducted to determine whether it was alcohol consumption or other adaptive stress coping behaviors (Appendix 3), and the results showed similar trends to those reported in past studies regarding adaptive stress coping and alcohol drinking behaviors after a disaster, we believe that the implications of the results obtained are meaningful. Second, most longitudinal studies suffer from imbalances in the number of repeated measures per individual, largely due to losses during follow-up (e.g., gender differences, workers being transferred due to radiation exposure, retirement, and refusal to participate further in the study). However, we confirmed that there were no significant differences in drinking alcohol in the presence or absence of two or more mental health follow-up time points, helping to exclude potential imbalances. Third, the participants in this study were all full-time TEPCO workers, and so it may not be possible to generalize our results to other populations. Female workers were also underrepresented in the present study, preventing us from performing gender-specific analyses, although gender was adjusted for in our models. Finally, although we adjusted for various possible confounding factors, residual confounding is

possible via presence of anxiety prior to the disaster; unmeasured variables, such as the intensity of radiation exposure; and social factors including socioeconomic or marital status.

The strengths of our study lie in its longitudinal design and repeated measures among workers who experienced a large-scale nuclear disaster. Two or more mental health symptoms measurement time points were available for most participants in the study (78.8% of the total 405 subjects in the study).

In summary, among the Fukushima nuclear power plant workers in this study, drinking alcohol for stress coping increased the risk of problem drinking, as well as PTSR and insomnia, especially in the absence of adaptive stress coping. These findings suggest that for an appropriate intervention, it is important to provide suitable education about effective self-coping and continuous monitoring, particularly alcohol drinking-related coping and its impact on mental health symptoms.

Declarations

Conflict of Interest Disclosures:

Shigemura provided voluntary mental health support to TEPCO Fukushima Daiichi and Daini nuclear power plant workers according to official requests from the Daini plant and a Japanese government cabinet order from the Ministry of Defense. Shigemura has also received consulting fees from the TEPCO Research Institute. Tanigawa is a part-time occupational physician at the Daini power plant. Tajima and Ikeda report no conflicts of interest.

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Author Contribution

Tajima, Ikeda, Shigemura, and Tanigawa conceptualized and developed the study design. Tanigawa and Shigemura collected the data. Tajima performed the statistical analyses. Tajima wrote the first draft of the manuscript. All authors contributed to the critical revision of the manuscript. All authors have read and approved the final manuscript.

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Data Availability

The data obtained in this research are retained only within specific computers with strict access control in the research participating organizations, as specified in the research protocol and approved by the relevant ethical review boards. These raw data will not be shared because only the outcomes of the analysis of the study population can be disclosed following the research protocol and commitment to the study participants specified in the Informed Consent Form.

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